Started on	Tuesday, 3 March 2026, 8:49 AM
State	Finished
Completed on	Tuesday, 3 March 2026, 8:51 AM
Time taken	2 mins 2 secs
Grade	**10.00** out of 100.00

Question 1

Partially correct

Mark 10.00 out of 15.00

Which of the following statements are true?

- a. (x > 0 || x < 10 && y < 0) is same as ((x > 0 || x < 10) && y < 0) ✖
- b. (x > 0 || x < 10) is same as ((x > 0) || (x < 10)) ✔
- c. (x > 0 && x < 10) is same as ((x > 0) && (x < 10)) ✔
- d. (x > 0 || x < 10 && y < 0) is same as (x > 0 || (x < 10 && y < 0))

Your answer is partially correct.

You have correctly selected 2.
The correct answers are:
(x > 0 && x < 10) is same as ((x > 0) && (x < 10)),

(x > 0 || x < 10) is same as ((x > 0) || (x < 10)),

(x > 0 || x < 10 && y < 0) is same as (x > 0 || (x < 10 && y < 0))

Question 2

Not answered

Mark 0.00 out of 15.00

Body Mass Index (BMI) is a measure of health based on height and weight. It can be calculated by taking your weight in kilograms and dividing it by the square of your height in meters. The interpretation of BMI for people 20 years or older is as follows:

BMI	Interpretation
BMI < 18.5	Underweight
18.5 ≤ BMI < 25.0	Normal
25.0 ≤ BMI < 30.0	Overweight
30.0 ≤ BMI	Obese

Write a JAVA program that prompts the user to enter a weight in pounds and height in inches and displays the BMI. Note that one pound is **0.45359237** kilograms and one inch is **0.0254** meters.

For example:

Input	Result
146 70	Enter weight in pounds: 146.0 Enter height in inches: 70.0 BMI is 20.948603801493316 Normal
175 55	Enter weight in pounds: 175.0 Enter height in inches: 55.0 BMI is 40.673446755984415 Obese

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
import java.util.Scanner;

public class ComputeAndInterpretBMI {

public static void main(String[] args) {
 Scanner input = new Scanner(System.in);

// Prompt the user to enter weight in pounds
System.out.print("Enter weight in pounds: ");

double weight = input.nextDouble();

System.out.println(weight);

// Prompt the user to enter height in inches
```

```
System.out.print("Enter height in inches: ");

double height = input.nextDouble();

System.out.println(height);

final double KILOGRAMS_PER_POUND = 0.45359237; // Constant

final double METERS_PER_INCH = 0.0254; // Constant


// Compute BMI

double weightInKilograms = weight * KILOGRAMS_PER_POUND;

double heightInMeters = height * METERS_PER_INCH;

double bmi = weightInKilograms / (heightInMeters * heightInMeters);


// Display result
System.out.println("BMI is " + bmi);

if (bmi < 18.5)
 System.out.println("Underweight");

else if (bmi < 25)
 System.out.println("Normal");

else if (bmi < 30)
 System.out.println("Overweight");

else
 System.out.println("Obese");
}
}
```

Question 3

Not answered

Mark 0.00 out of 15.00

Write a JAVA program that prompts the user to enter an integer for today's day of the week (Sunday is 0, Monday is 1, ..., and Saturday is 6).

Also prompt the user to enter the number of days after today for a future day and display the future day of the week.

For example:

Input	Result
1 3	Enter today's day: 1 Enter the number of days elapsed since today: 3 Today is Monday and the future day is Thursday
0 31	Enter today's day: 0 Enter the number of days elapsed since today: 31 Today is Sunday and the future day is Wednesday

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
public class Exercise03_05 {
  public static void main(String[] args) {
    java.util.Scanner input = new java.util.Scanner(System.in);

    // Prompt the user to enter an integer for today
    System.out.print("Enter today's day: ");
    int today = input.nextInt();
    System.out.println(today);

    System.out.print("Enter the number of days elapsed since today: ");
    int elapsedDays = input.nextInt();
    System.out.println(elapsedDays);

    String nameForToday;
    if (today == 0)
      nameForToday = "Sunday";
    else if (today == 1)
      nameForToday = "Monday";
    else if (today == 2)
      nameForToday = "Tuesday";
    else if (today == 3)
      nameForToday = "Wednesday";
    else if (today == 4)
      nameForToday = "Thursday";
    else if (today == 5)
      nameForToday = "Friday";
    else // if (today == 6)
      nameForToday = "Saturday";
```

```
29
30      int futureDay = (today + elapsedDays) % 7;
31      String nameForFutureDay;
32      if (futureDay == 0)
33        nameForFutureDay = "Sunday";
34      else if (futureDay == 1)
35        nameForFutureDay = "Monday";
36      else if (futureDay == 2)
37        nameForFutureDay = "Tuesday";
38      else if (futureDay == 3)
39        nameForFutureDay = "Wednesday";
40      else if (futureDay == 4)
41        nameForFutureDay = "Thursday";
42      else if (futureDay == 5)
43        nameForFutureDay = "Friday";
44      else // if (futureDay == 6)
45        nameForFutureDay = "Saturday";
46
47      System.out.println("Today is " + nameForToday
48          + " and the future day is " + nameForFutureDay);
49    }
50  }
51
```

Question 4

Not answered

Mark 0.00 out of 15.00

Write a JAVA program that prompts the user to enter a three-digit integer and determines whether it is a palindrome number.

Also you should first check whether the input is valid or not.

A number is palindrome if it reads the same from right to left and from left to right.

Consider the following examples:

For example:

Input	Result
121	Enter a three-digit integer: 121 121 is a palindrome
18745	Enter a three-digit integer: 18745 Invalid Entry
546	Enter a three-digit integer: 546 546 is not a palindrome

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
import java.util.Scanner;

public class Exercise03_12 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);

    System.out.print("Enter a three-digit integer: ");
    int number = input.nextInt();
    System.out.println(number);

    if (number > 999 || number < 100){
        System.out.println("Invalid Entry");
    }
    else{
        if (number / 100 == number % 10)
            System.out.println(number + " is a palindrome");
        else
            System.out.println(number + " is not a palindrome");
    }
  }
}
```

Question 5

Not answered

Mark 0.00 out of 20.00

Write a JAVA program that plays the popular scissor–rock–paper game. (A scissor can cut a paper, a rock can knock a scissor, anda paper can wrap a rock.)

The program randomly generates a number 0, 1, or 2 representing scissor, rock, and paper.

The program prompts the user to enter a number 0, 1, or 2 and displays a message indicating whether the user or the computer wins, loses, or draws.

Hint: The **java.lang.Math.random()** method returns a pseudorandom double type number greater than or equal to 0.0 and less than 1.0.

For example:

Input	Result
1	scissor (0), rock (1), paper (2): 1 The computer is paper. You are rock. You lost

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
public class Exercise03_17 {
  public static void main(String[] args) {
    // Generate scissor, rock, paper
    int computerNumber = 2;//(int)(Math.random() * 3);

    // Prompt the user to enter scissor, rock, or paper
    java.util.Scanner input = new java.util.Scanner(System.in);
    System.out.print("scissor (0), rock (1), paper (2): ");
    int userNumber = input.nextInt();
    System.out.println(userNumber);
    // Check the guess
    switch (computerNumber) {
      case 0:
        if (userNumber == 0)
          System.out.print("The computer is scissor. You are scissor too. It is a draw");
        else if (userNumber == 1)
          System.out.print("The computer is scissor. You are rock. You won");
        else if (userNumber == 2)
          System.out.print("The computer is scissor. You are paper. You lost");
        break;
      case 1:
        if (userNumber == 0)
          System.out.print("The computer is rock. You are scissor. You lost");
        else if (userNumber == 1)
          System.out.print("The computer is rock. You are rock too. It is a draw");
```

```
        else if (userNumber == 2)
          System.out.print("The computer is rock. You are paper. You won");
        break;
      case 2:
        if (userNumber == 0)
          System.out.print("The computer is paper. You are scissor. You won");
        else if (userNumber == 1)
          System.out.print("The computer is paper. You are rock. You lost");
        else if (userNumber == 2)
          System.out.print("The computer is paper. You are paper too. It is a draw");
        break;
       }
    }
}
```

Question 6

Not answered

Mark 0.00 out of 20.00

Write a JAVA program that reads three edges for a triangle and computes the perimeter if the input is valid.

Otherwise, display that the input is invalid. The input is valid if the sum of every pair of two edges is greater than the remaining edge.

For example:

Input	Result
5 12 13	Enter three edges (length in double): 5.0 , 12.0 , 13.0 The perimeter is 30.0
5 12 3	Enter three edges (length in double): 5.0 , 12.0 , 3.0 Input is invalid

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
public class Exercise03_19 {
  public static void main(String[] args) {
    java.util.Scanner input = new java.util.Scanner(System.in);

    // Enter three edges
    System.out.print(
      "Enter three edges (length in double): ");
    double edge1 = input.nextDouble();
    double edge2 = input.nextDouble();
    double edge3 = input.nextDouble();
    System.out.println(edge1 + " , " + edge2 + " , " + edge3);

    // Display results
    boolean isValid = (edge1 + edge2 > edge3) &&
      (edge1 + edge3 > edge2) && (edge2 + edge3 > edge1);

    // Display results
    if (isValid) {
      System.out.println("The perimeter is " + (edge1 + edge2 + edge3));
    }
    else {
      System.out.println("Input is invalid");
    }
  }
}

```